

Uncertainty Quantification in Deep Learning for Histopathology: When Should the Model Say "I Don't Know"?

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ABSTRACT

Clinical deployment of diagnostic AI requires not just accurate predictions but calibrated uncertainty — a model should express low confidence when it is likely to be wrong. We systematically evaluate six uncertainty quantification (UQ) methods across three histopathology classification tasks (breast cancer grading, prostate cancer Gleason scoring, lung adenocarcinoma subtyping) using a multi-site dataset of 28,000 whole-slide images. Monte Carlo Dropout achieves the best calibration (Expected Calibration Error 0.043) while Deep Ensembles achieve the highest AUROC for uncertainty-guided abstention (0.87 — identifying cases where the model should defer to a pathologist). We demonstrate a human-AI collaborative workflow where AI abstains on 18% of cases, increasing overall system accuracy from 88.2% to 94.7% with no additional pathologist workload.

METHODS

- Dataset: 28,000 WSIs from 5 pathology labs; 3 tasks (breast, prostate, lung)
- UQ methods: MC Dropout, Deep Ensembles, Evidential Deep Learning, Laplace Approximation, Conformal Prediction, Platt Scaling
- Base architecture: CONCH (pathology foundation model) + task heads
- Metrics: Expected Calibration Error, AUROC for abstention, Brier Score
- Human-AI workflow simulation: pathologist review triggered by model uncertainty score > 0.7

KEY FINDINGS

1. Deep Ensembles best for uncertainty-guided abstention (AUROC 0.87)
2. MC Dropout best calibration (ECE 0.043) with lowest compute overhead
3. Selective prediction (AI + abstention) achieves 94.7% accuracy vs 88.2% for AI alone
4. Conformal prediction provides rigorous coverage guarantees — practically useful for regulatory submissions
5. Models most uncertain on: image artefacts, rare subtypes, edge-of-grade-boundary cases (clinically meaningful)

LIMITATIONS

- Computational cost of Deep Ensembles (5 forward passes) may be prohibitive in high-throughput settings
- UQ methods calibrated on in-distribution data; out-of-distribution behaviour not fully characterised
- Human-AI workflow simulation uses synthetic pathologist response — not prospectively validated

RELEVANCE TO ML IN CLINICAL DIAGNOSTICS: HIGH

Directly addresses regulatory and clinical trust requirements for AI deployment. The human-AI collaborative framework is a practical contribution for system design.